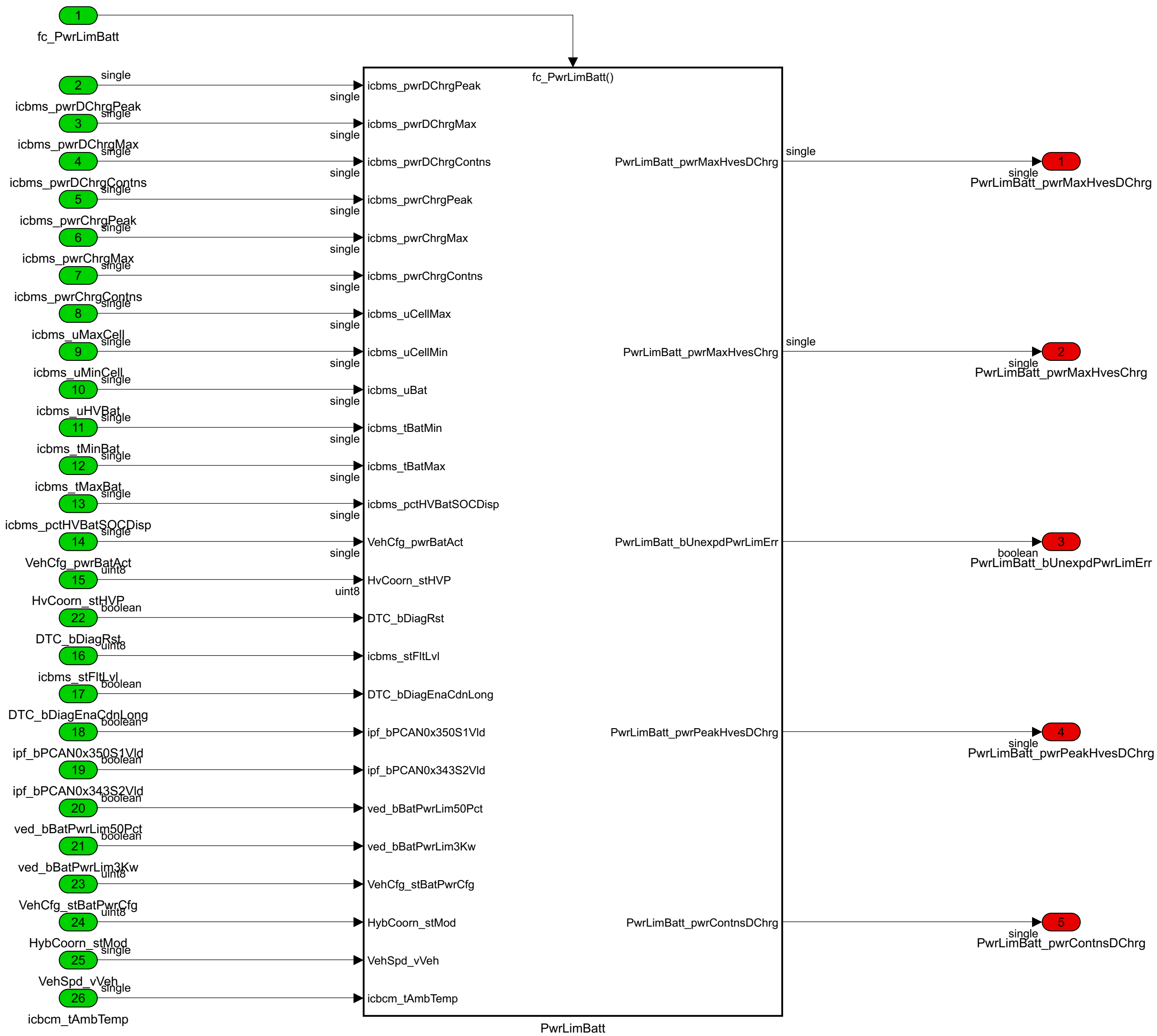
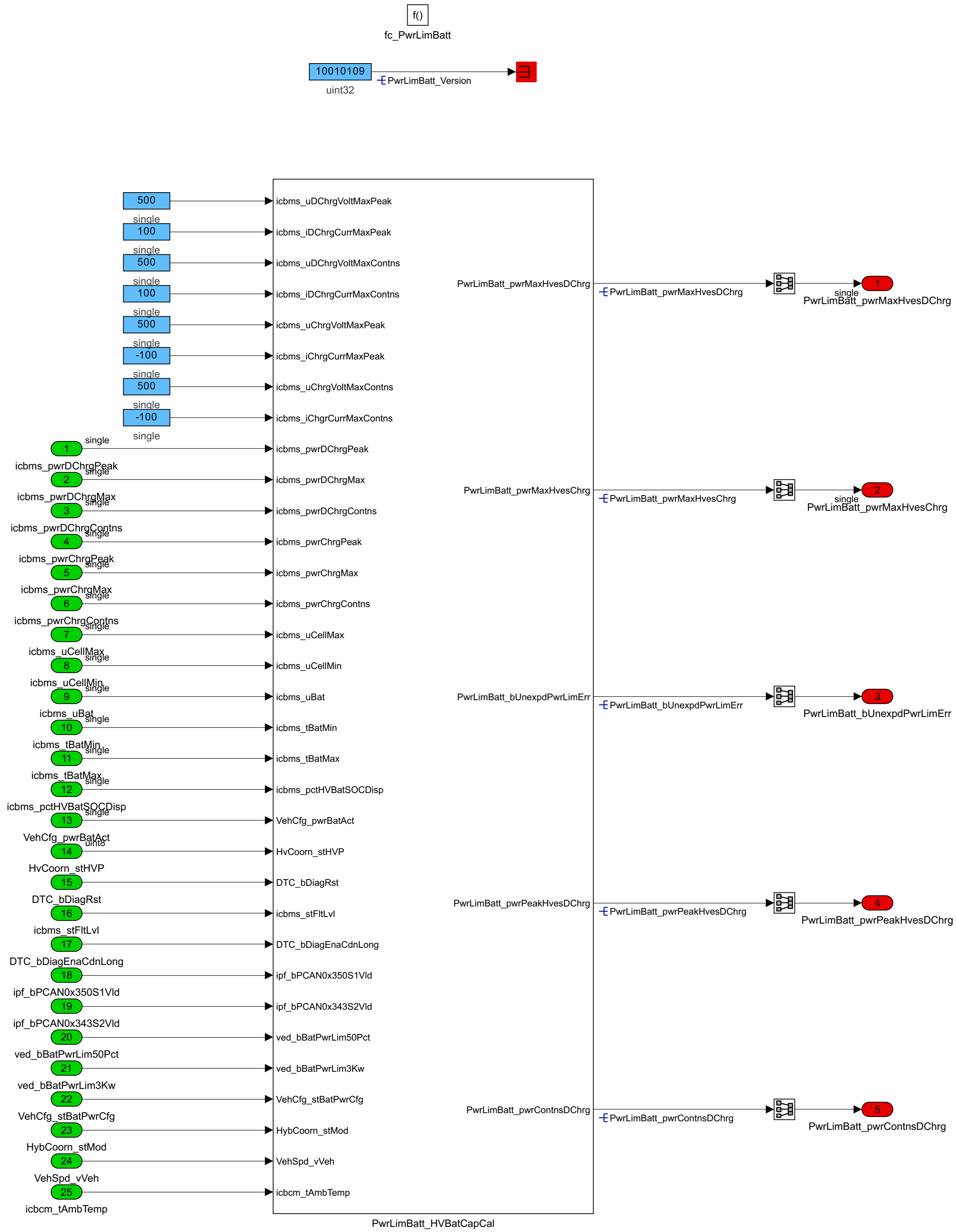
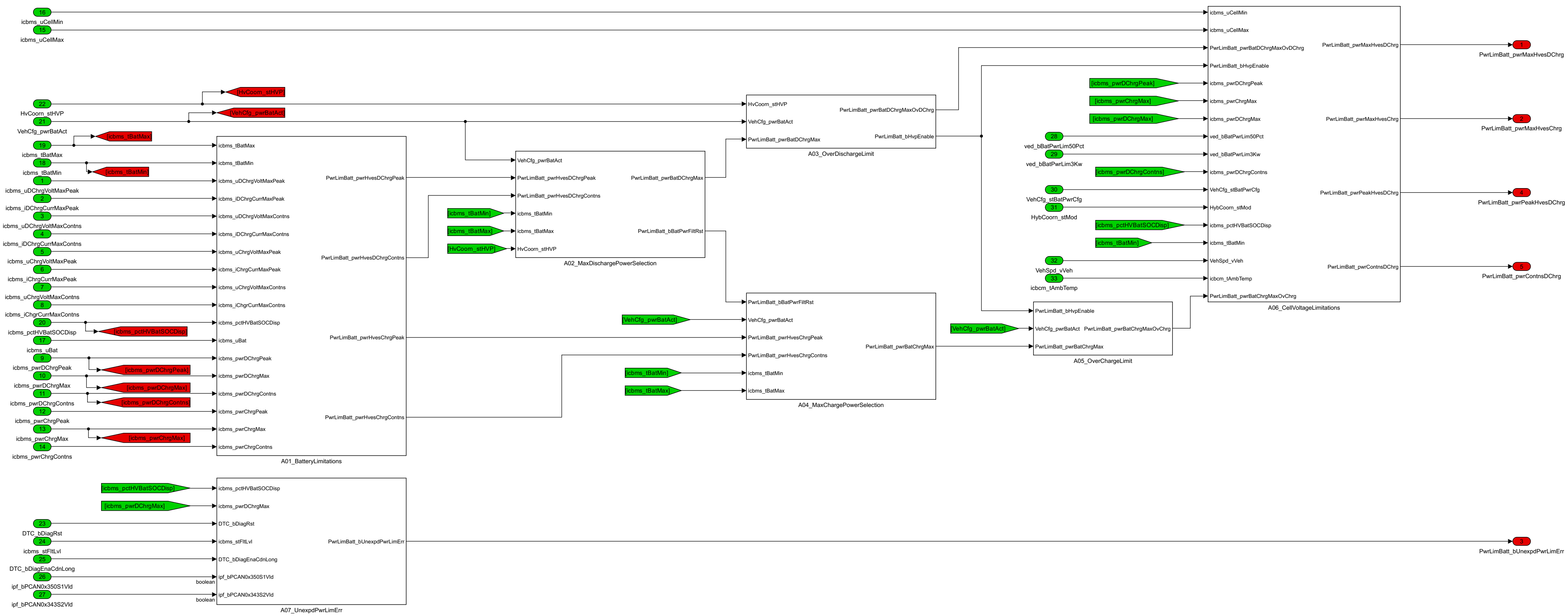
<ModelName>: PwrLimBatt
<Project>: Chery_ICAR05
<Creator>: LiShuai
<CreatedDate>: 2022 08 02 16:50
<ModelVersion>: 1.1.1.9
<Description>: High Voltage Battery Power Limit
xuliu1
Tue Mar 10 14:41:37 2026



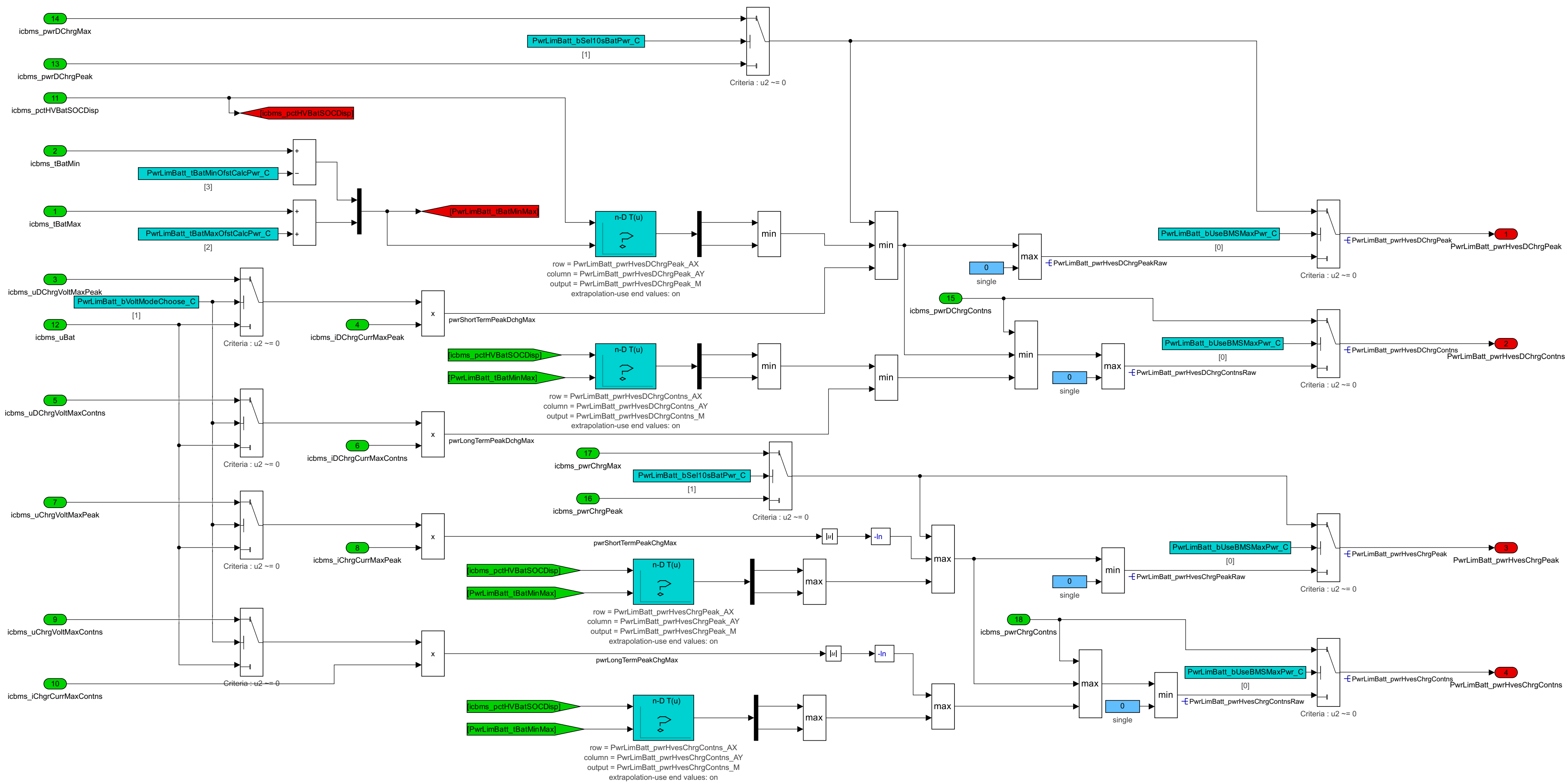


f()
fc_PwrLimBatt
10010109
uint32
PwrLimBatt_Version
500
100
500
100
500
-100
500
-100
single
icbms_uDChrgVoltMaxPeak
icbms_iDChrgCurrMaxPeak
icbms_uDChrgVoltMaxContns
icbms_iDChrgCurrMaxContns
icbms_uChrgVoltMaxPeak
icbms_iChrgCurrMaxPeak
icbms_uChrgVoltMaxContns
icbms_iChgrCurrMaxContns
icbms_pwrDChrgPeak
icbms_pwrDChrgMax
icbms_pwrDChrgContns
icbms_pwrChrgPeak
icbms_pwrChrgMax
icbms_pwrChrgContns
icbms_uCellMax
icbms_uCellMin
icbms_uBat
icbms_tBatMin
icbms_tBatMax
icbms_pctHVBatSOCDisp
VehCfg_pwrBatAct
uint8
HvCoorn_stHVP
DTC_bDiagRst
icbms_stFltLvl
DTC_bDiagEnaCdnLong
ipf_bPCAN0x350S1Vld
ipf_bPCAN0x343S2Vld
ved_bBatPwrLim50Pct
ved_bBatPwrLim3Kw
VehCfg_stBatPwrCfg
HybCoorn_stMod
VehSpd_vVeh
icbcm_tAmbTemp
PwrLimBatt_HVBatCapCal
PwrLimBatt_pwrMaxHvesDChrg
PwrLimBatt_pwrMaxHvesChrg
PwrLimBatt_bUnexpdPwrLimErr
PwrLimBatt_pwrPeakHvesDChrg
PwrLimBatt_pwrContnsDChrg

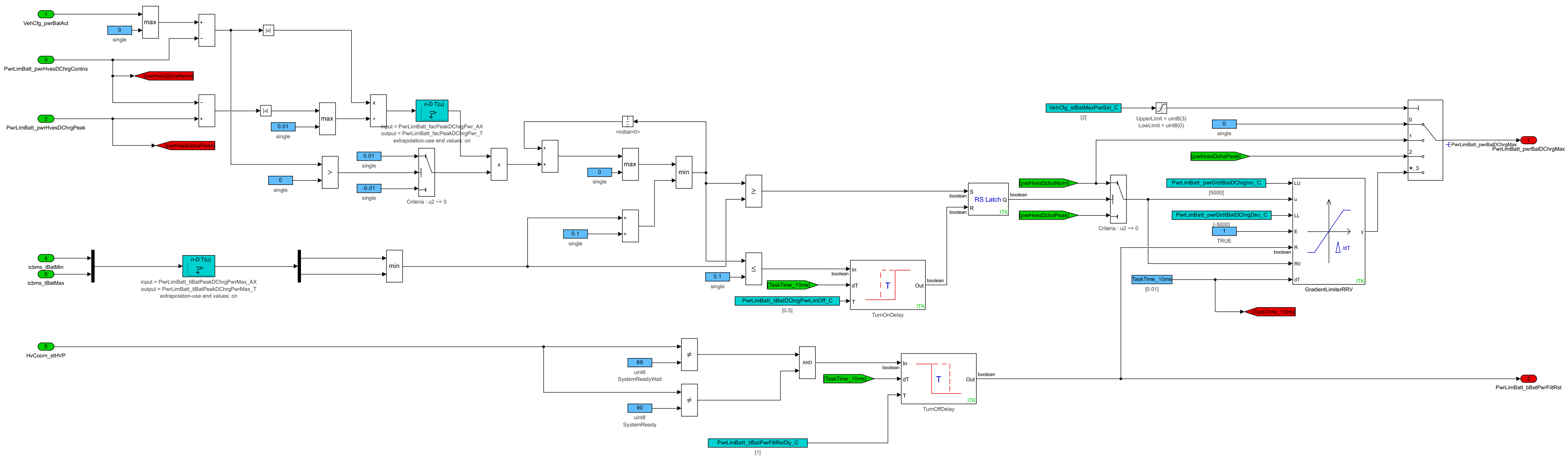


icbms_uCellMin
icbms_uCellMax
HvCoorn_stHVP
VehCfg_pwrBatAct
icbms_tBatMax
icbms_tBatMin
icbms_uDChrgVoltMaxPeak
icbms_iDChrgCurrMaxPeak
icbms_uDChrgVoltMaxContns
icbms_iDChrgCurrMaxContns
icbms_uChrgVoltMaxPeak
icbms_iChrgCurrMaxPeak
icbms_uChrgVoltMaxContns
icbms_iChgrCurrMaxContns
icbms_pctHVBatSOCDisp
icbms_uBat
icbms_pwrDChrgPeak
icbms_pwrDChrgMax
icbms_pwrDChrgContns
icbms_pwrChrgPeak
icbms_pwrChrgMax
icbms_pwrChrgContns
A01_BatteryLimitations
PwrLimBatt_pwrHvesDChrgPeak
PwrLimBatt_pwrHvesDChrgContns
PwrLimBatt_pwrHvesChrgPeak
PwrLimBatt_pwrHvesChrgContns
A02_MaxDischargePowerSelection
PwrLimBatt_pwrBatDChrgMax
PwrLimBatt_bBatPwrFiltRst
A03_OverDischargeLimit
PwrLimBatt_pwrBatDChrgMaxOvDChrg
PwrLimBatt_bHvpEnable
A04_MaxChargePowerSelection
PwrLimBatt_pwrBatChrgMax
A05_OverChargeLimit
PwrLimBatt_pwrBatChrgMaxOvChrg
A06_CellVoltageLimitations
ved_bBatPwrLim50Pct
ved_bBatPwrLim3Kw
VehCfg_stBatPwrCfg
HybCoorn_stMod
VehSpd_vVeh
icbcm_tAmbTemp
PwrLimBatt_pwrMaxHvesDChrg
PwrLimBatt_pwrMaxHvesChrg
PwrLimBatt_pwrPeakHvesDChrg
PwrLimBatt_pwrContnsDChrg
A07_UnexpdPwrLimErr
DTC_bDiagRst
icbms_stFltLvl
DTC_bDiagEnaCdnLong
ipf_bPCAN0x350S1Vld
ipf_bPCAN0x343S2Vld
boolean
PwrLimBatt_bUnexpdPwrLimErr

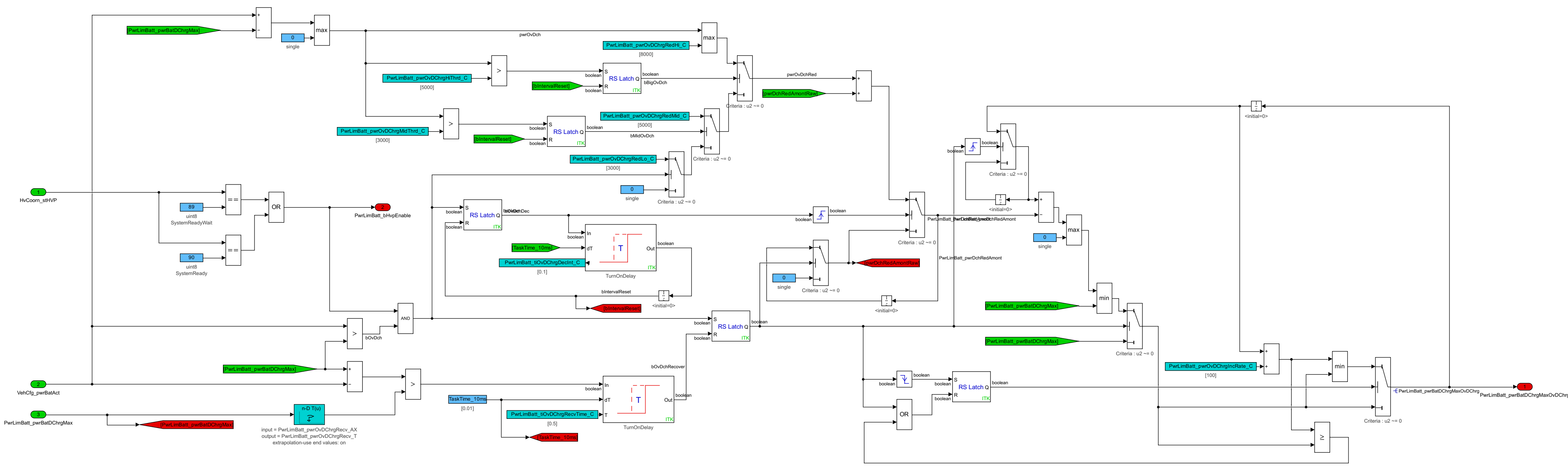
This subsystem is mainly used to calculate the peak and continuous discharge and charge power of HV system, based on the peak discharge and charge current,the peak discharge and charge voltage, the continuous discharge and charge current,the continuous discharge and charge voltage, batteruy actual soc, the maximum and minimum temperature of Hv battery, etc.



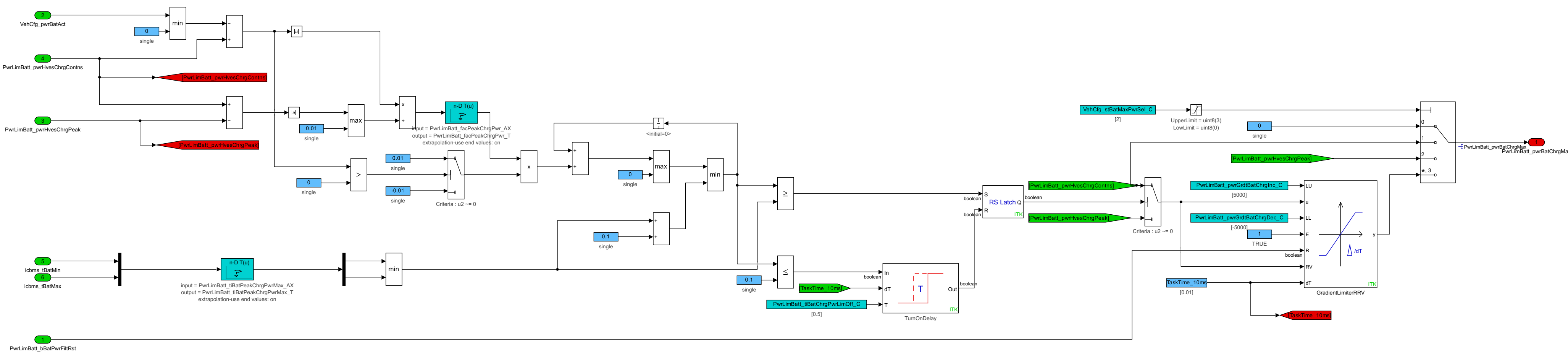
This subsystem is mainly used to calculate the maximum discharge power of HV battery, based on the peak discharg power,the continuous discharge power, the actual power of battery, the maximum and minimum temperature of Hv battery, etc.



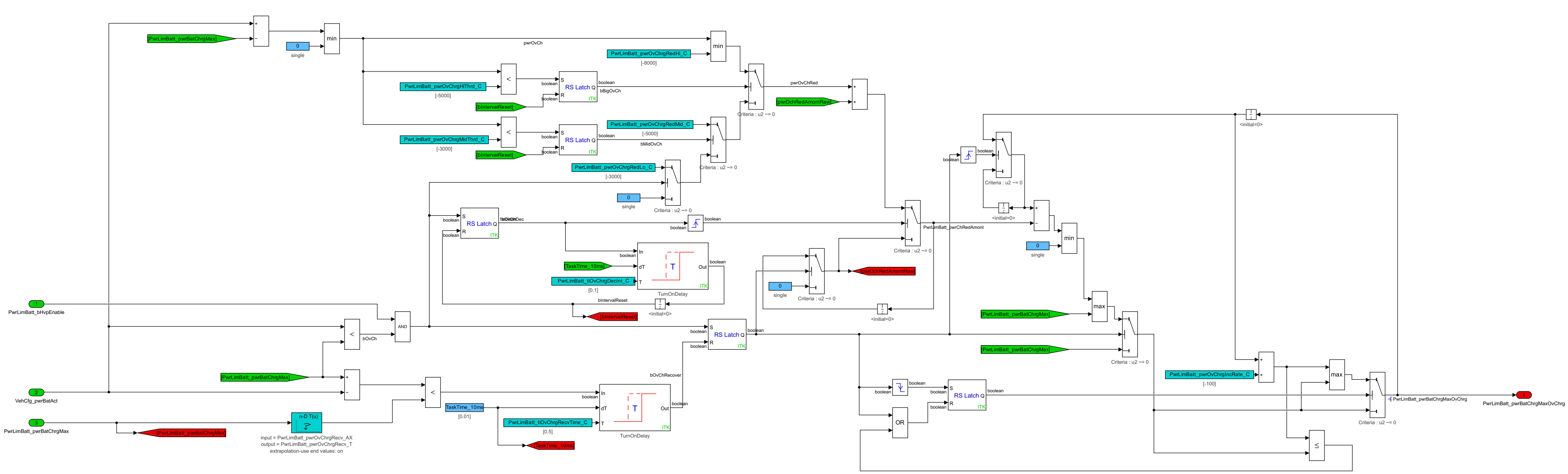
This subsystem is mainly used to calculate the maximum discharge power of HV battery after battery over discharge control, based on the maximum discharg power, the actual power of battery, etc.
When the actual power is larger than the maximum discharge power of battery, the maximum discharge power of HV battery after battery over discharge control will decrease gradually to 0.



This subsystem is mainly used to calculate the maximum charge power of HV battery, based on the peak charg power,the continuous charge power, the actual power of battery, the maximum and minimum temperature of Hv battery, etc.



This subsystem is mainly used to calculate the maximum charge power of HV battery after battery over charge control, based on the maximum charg power, the actual power of battery, etc.
When the actual power is larger than the maximum charge power of battery, the maximum charge power of HV battery after battery over charge control will decrease gradually to 0.



Documentation

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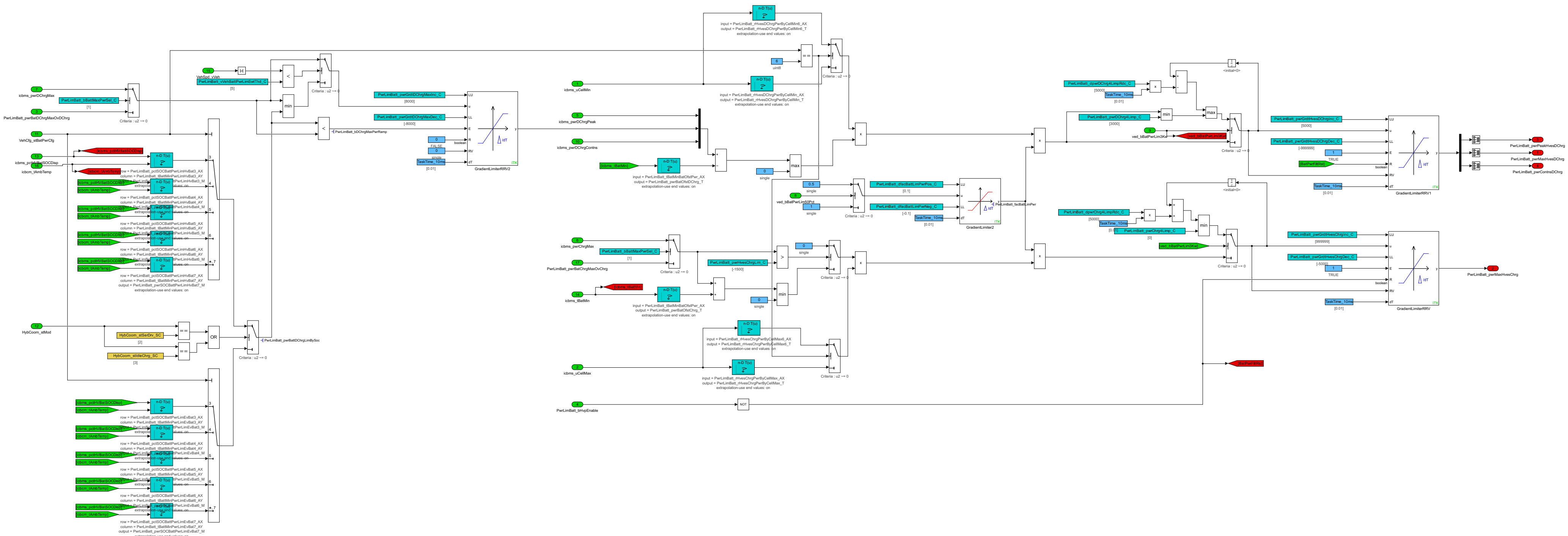
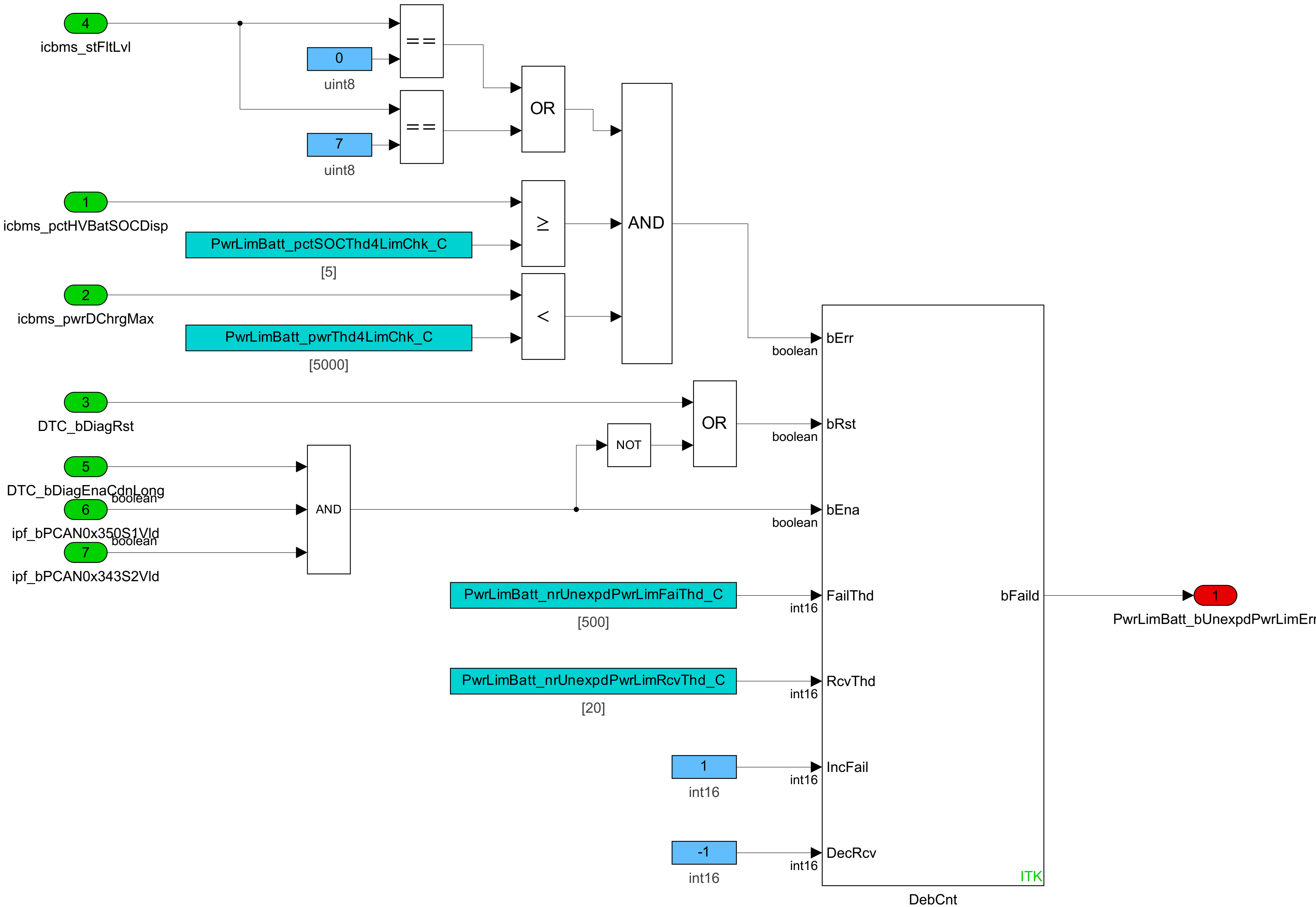
This bolck limit the battery output power by the cell voltage. However, when the battery charge capability is lower than bcc_pwrHvesChgLi
Finally, the batter discharge or charge power is filtered by a low pass filter to pervent power fluctuation.

DOC
Text
Unit detailed design





4
icbms_stFltLvl
0
uint8
==
7
uint8
==
OR
1
icbms_pctHVBatSOCDisp
PwrLimBatt_pctSOCThd4LimChk_C
[5]
≥
2
icbms_pwrDChrgMax
PwrLimBatt_pwrThd4LimChk_C
[5000]
<
AND
3
DTC_bDiagRst
5
DTC_bDiagEnaCdnLong
boolean
6
ipf_bPCAN0x350S1Vld
boolean
7
ipf_bPCAN0x343S2Vld
AND
NOT
OR
bErr
boolean
bRst
boolean
bEna
boolean
PwrLimBatt_nrUnexpdPwrLimFaiThd_C
[500]
FailThd
int16
PwrLimBatt_nrUnexpdPwrLimRcvThd_C
[20]
RcvThd
int16
1
int16
IncFail
int16
-1
int16
DecRcv
int16
bFaild
ITK
DebCnt
1
PwrLimBatt_bUnexpdPwrLimErr